1. BEMAC Class B (non-corner, symmetric rate)

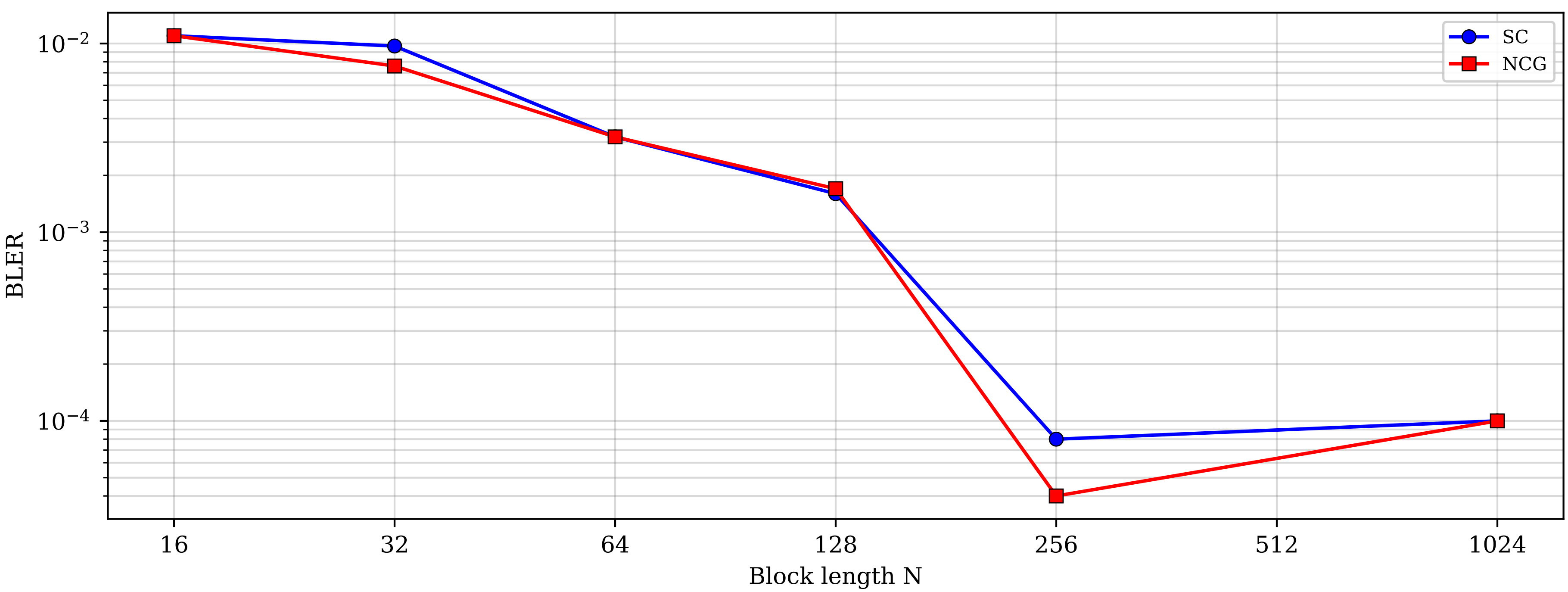
Binary Erasure MAC -- Z = X + Y

`Z = X + Y,  Z in {0, 1, 2}  (deterministic)`

Path: make_path(N, N//2) = [0]^{N/2} [1]^N [0]^{N/2}
Capacity: I(X;Z)=0.500, I(Y;Z|X)=1.000, Symmetric point: (0.750, 0.750)
Operating rate: R_U ~ 0.50, R_V ~ 0.70
Architecture: NCG d=16, hidden=64, BEMAC vocab=3 embedding

N	ku	kv	SC BLER	SC errs/CW	NCG BLER	CRC-SCL L=4
16	8	11	0.0110	~55/5k	0.0110	--
32	16	22	0.0097	146/15k	0.0076	0.0000
64	32	44	0.0032	128/40k	0.0032	0.0000
128	64	89	0.0016	81/50k	0.0017	0.0000
256	128	178	8.0e-05	~4/50k	4.0e-05	0.0000
512	256	358	0.0000	0/2k	0.0000	--
1024	512	716	1.0e-04	~1/10k	1.0e-04	--

BLER vs N -- NCG ~ SC on Class B. CRC-SCL L=4 achieves zero errors at all tested N.



2. BEMAC Class C (corner rate)

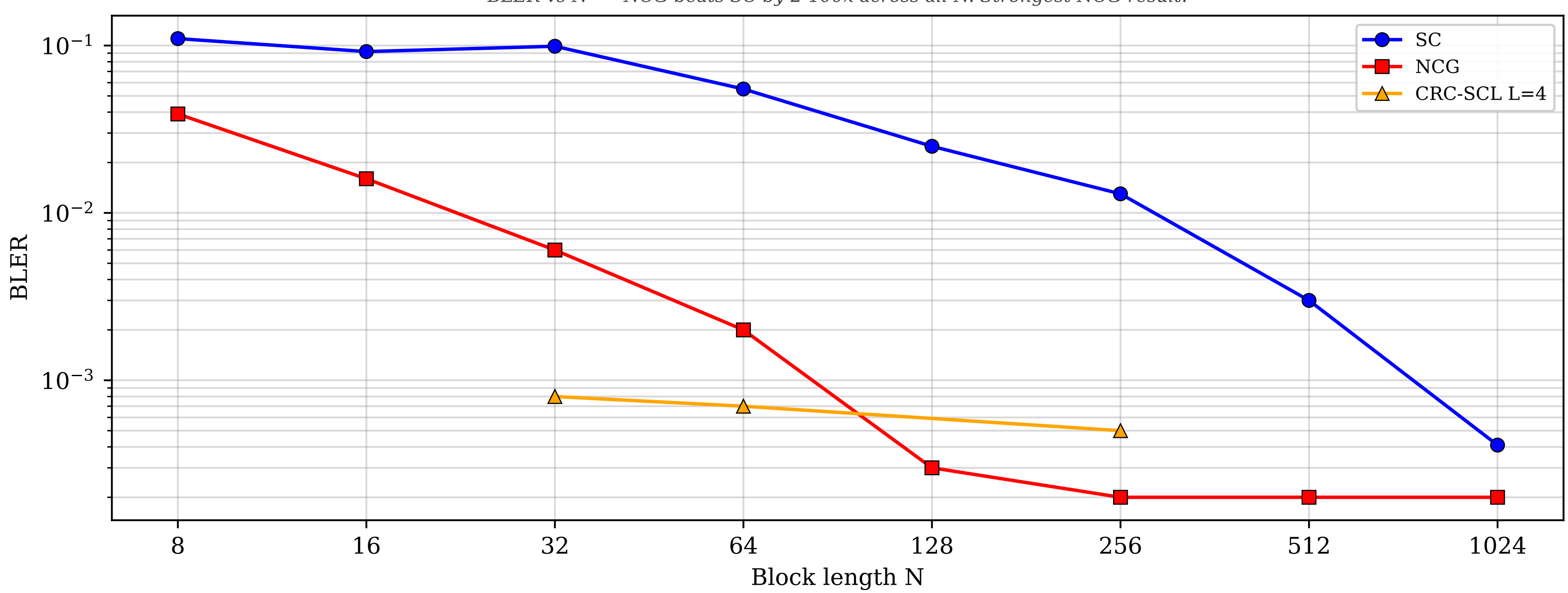
Binary Erasure MAC -- $Z = X + Y$

`Z = X + Y,  Z in {0, 1, 2}  (deterministic)`

Path: make_path(N, N) = [0]^N [1]^N (all U first, then all V)
Capacity: Corner point (R_U, R_V) = (0.500, 1.000)
Operating rate: R_U ~ 0.30, R_V ~ 0.60 (60% of corner capacity)
Architecture: NCG d=16, hidden=64, BEMAC vocab=3 embedding

N	ku	kv	SC BLER	SC errs/CW	NCG BLER	NCG/SC	CRC-SCL L=4
8	2	5	0.1100	329/3k	0.0390	0.36x	--
16	5	10	0.0920	276/3k	0.0160	0.18x	--
32	10	19	0.0990	298/3k	0.0060	0.06x	8.0e-04
64	19	38	0.0550	165/3k	0.0020	0.03x	7.0e-04
128	38	77	0.0250	491/20k	3.0e-04	0.01x	0.0000
256	77	154	0.0130	668/50k	2.0e-04	0.01x	5.0e-04
512	154	307	0.0030	168/50k	2.0e-04	0.06x	--
1024	307	614	4.1e-04	26/64k	2.0e-04	0.42x	--

BLER vs N -- NCG beats SC by 2-100x across all N. Strongest NCG result.



3. GMAC Class B (non-corner, symmetric rate)

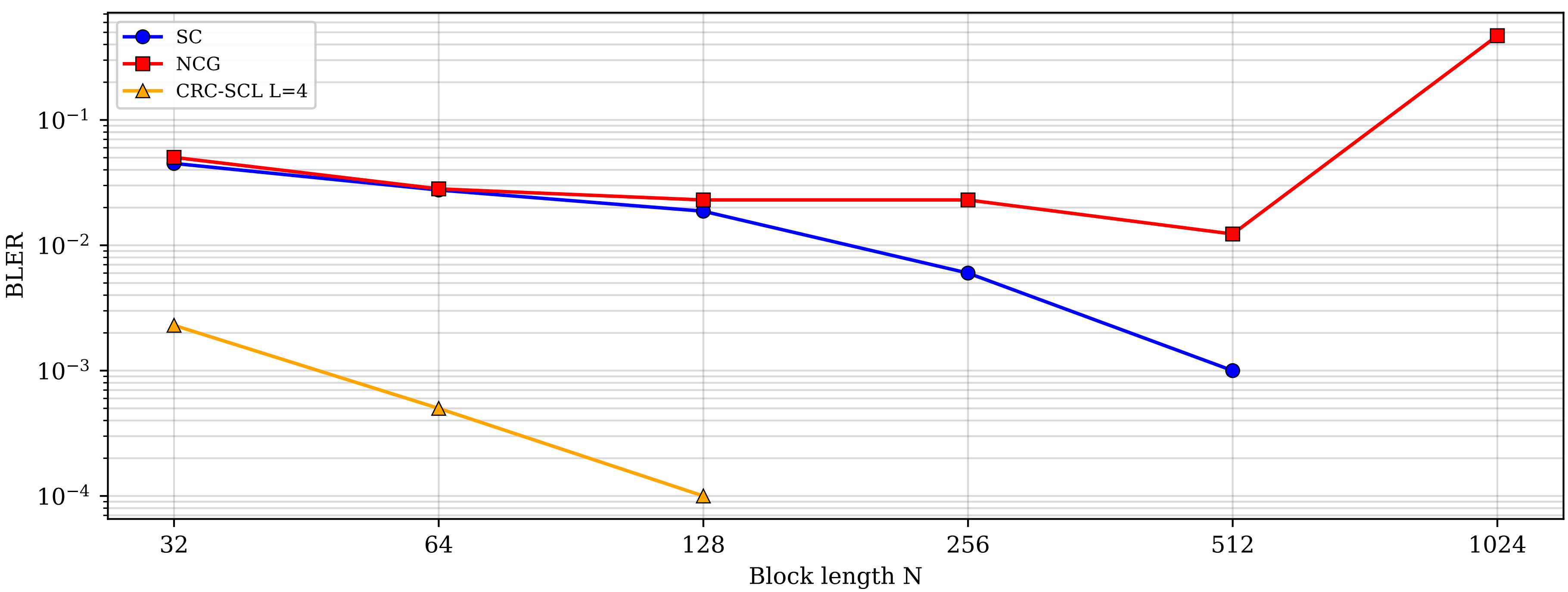
Gaussian MAC -- Z = (1-2X) + (1-2Y) + W

`Z = (1-2X) + (1-2Y) + W,  W ~ N(0, sigma^2),  SNR = 6 dB`

Path: make_path(N, N//2), symmetric rate ~(0.688, 0.688)
Capacity: I(X;Z)=0.465, I(Y;Z|X)=0.912, I(X,Y;Z)=1.376
Operating rate: R_U = R_V ~ 0.48 (~70% of symmetric capacity)
Architecture: NCG d=16, hidden=64, GMAC z_encoder MLP

N	ku=kv	SC BLER	SC errs/CW	NCG BLER	NCG/SC	CRC-SCL L=4
32	15	0.0450	135/3k	0.0503	1.12x	0.0023
64	31	0.0276	138/5k	0.0282	1.02x	5.0e-04
128	62	0.0187	112/6k	0.0230	1.23x	1.0e-04
256	123	0.0060	30/5k	0.0230	4.5x	0.0000
512	246	0.0010	est.	0.0123	12.3x	0.0000
1024	492	--	--	0.4700	BROKEN	--

BLER vs N -- NCG does NOT beat SC on GMAC Class B. Wall at N=256. CRC-SCL L=4 is the dominant decoder.



4. GMAC Class C (corner rate)

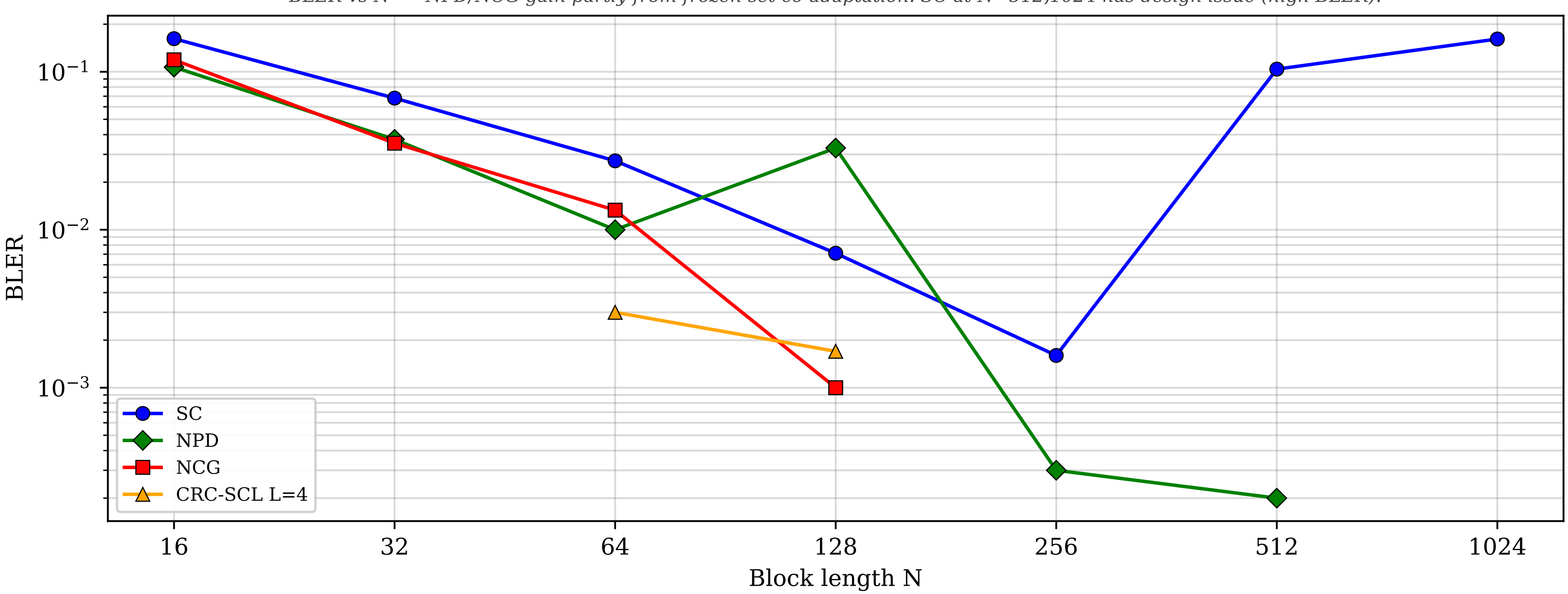
Gaussian MAC -- Z = (1-2X) + (1-2Y) + W

`Z = (1-2X) + (1-2Y) + W,  W ~ N(0, sigma^2),  SNR = 6 dB`

Path: make_path(N, N), corner rate (R_U ~ 0.23, R_V ~ 0.45)
Capacity: Corner point (R_U, R_V) = (0.465, 0.912)
NPD/NCG use MI-designed frozen sets (co-adapted with NN decoder)
Architecture: NPD d=16, hidden=64, neural fast_ce (use_analytical=False)

N	ku	kv	SC BLER	SC errs/CW	NPD BLER	NCG BLER	CRC-SCL L=4
16	4	7	0.1620	1620/10k	0.1070	0.1190	--
32	7	15	0.0681	681/10k	0.0373	0.0353	--
64	15	29	0.0273	273/10k	0.0100	0.0133	0.0030
128	30	58	0.0071	71/10k	0.0329	0.0010	0.0017
256	59	117	0.0016	31/20k	3.0e-04	--	0.0000
512	119	233	0.1039	5197/50k	2.0e-04	--	--
1024	238	467	0.1612	8058/50k	0.0000	--	--

BLER vs N -- NPD/NCG gain partly from frozen-set co-adaptation. SC at N=512,1024 has design issue (high BLER).



5. ABNMAC Class B (non-corner, symmetric rate)

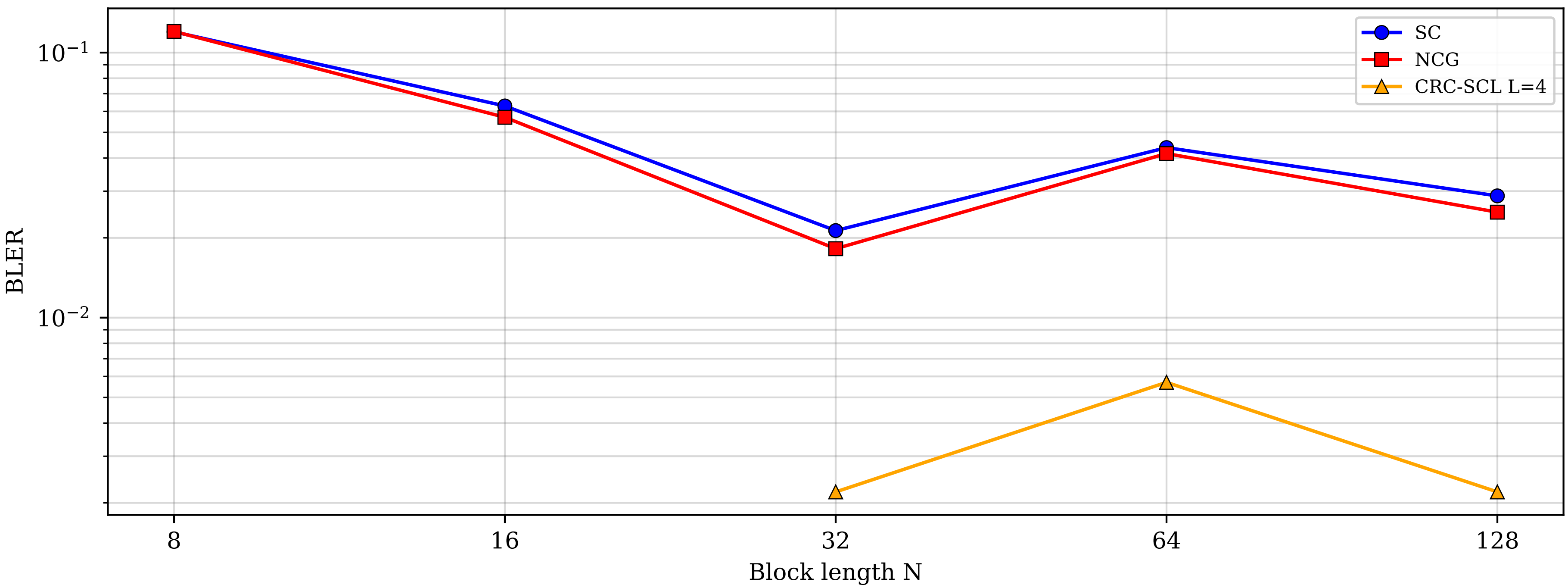
Asymmetric Binary Noise MAC

`Z = (X xor Ex, Y xor Ey),  correlated binary noise matrix`

Path: make_path(N, N//2), symmetric rate ~(0.600, 0.600)
Capacity: I(X;Z) ~ 0.400, I(Y;Z|X) ~ 0.800, Symmetric point: (0.600, 0.600)
Operating rate: R_U ~ R_V ~ 0.30 (ku=kv)
Architecture: NCG d=16, hidden=64

N	ku=kv	SC BLER	SC errs/CW	NCG BLER	NCG/SC	CRC-SCL L=4
8	3	0.1198	1198/10k	0.1202	1.00x	--
16	5	0.0629	629/10k	0.0570	0.91x	--
32	10	0.0213	213/10k	0.0182	0.85x	0.0022
64	22	0.0438	219/5k	0.0416	0.95x	0.0057
128	45	0.0288	115/4k	0.0250	--	0.0022

BLER vs N -- NCG gains are modest (0-15%). Non-monotonic SC BLER at N=64 > N=32 (rate-selection artifact).



6. ABNMAC Class C (corner rate)

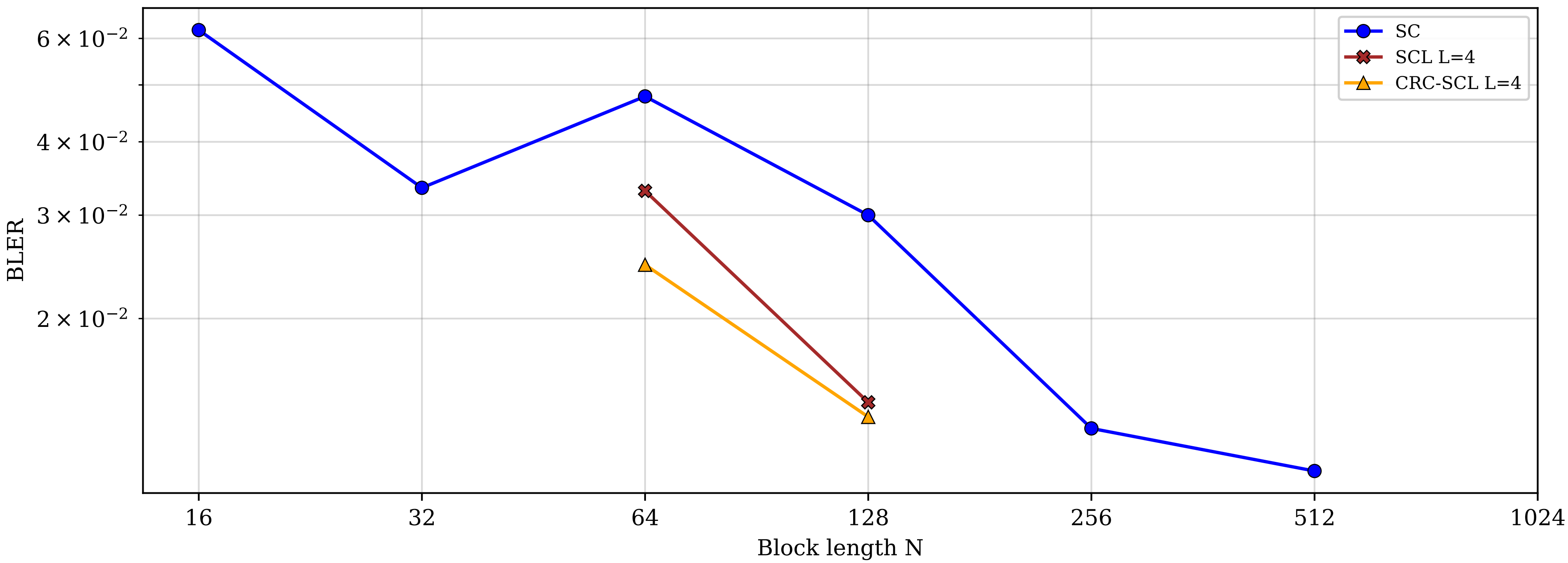
Asymmetric Binary Noise MAC -- SC baseline only

`Z = (X xor Ex, Y xor Ey),  correlated binary noise matrix`

Path: make_path(N, N) = [0]^N [1]^N
Capacity: Corner point (R_U, R_V) = (0.400, 0.800)
Operating rate: R_U ~ 0.19, R_V ~ 0.38
No neural decoder trained (ABNMAC tuple output incompatible with current z_encoder)

N	ku	kv	SC BLER	SC errs/CW	SCL L=4	CRC-SCL L=4
16	3	6	0.0620	?/?	--	--
32	6	13	0.0334	167/5k	--	--
64	13	26	0.0478	239/5k	0.0330	0.0247
128	26	51	0.0300	150/5k	0.0144	0.0136
256	51	102	0.0130	?/?	--	--
512	102	205	0.0110	?/?	--	--
1024	205	410	0.0000	?/?	--	--

BLER vs N -- No neural decoder tested. Non-monotonic SC BLER confirmed (N=64 > N=32). CRC-SCL gives 2-3x at N=64,128.



7. ISI-MAC Class C (corner rate) -- Memory Channel

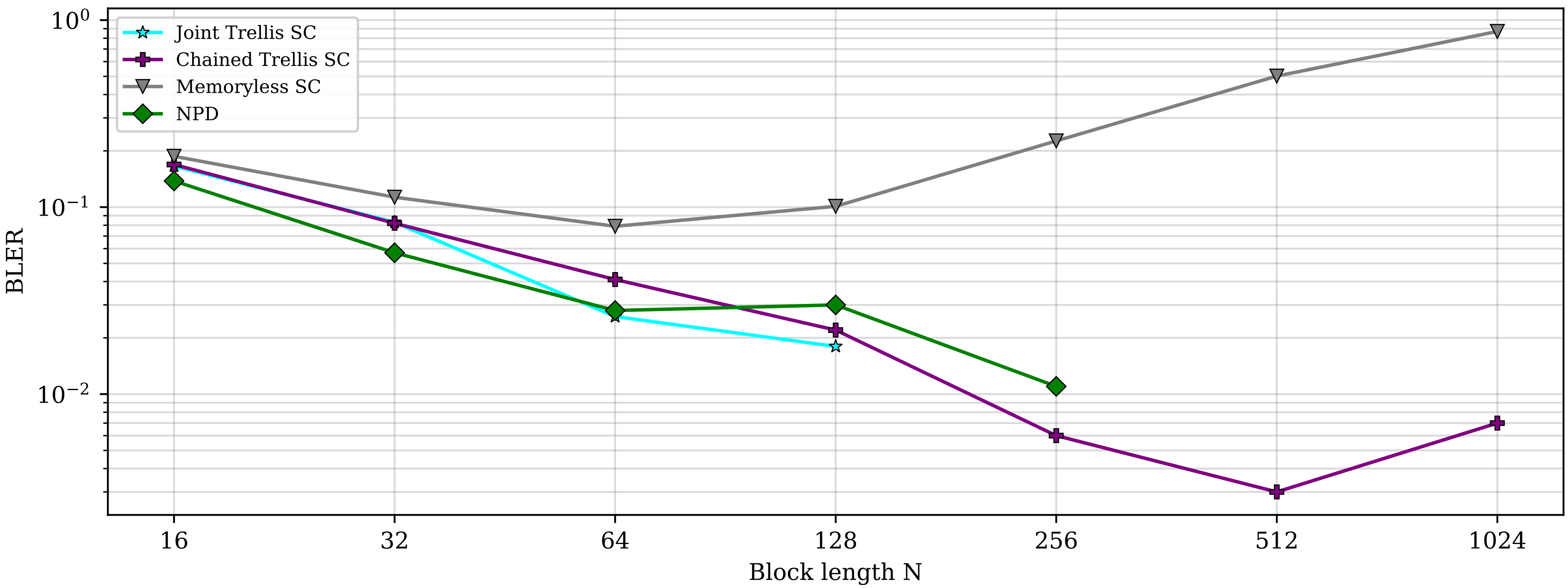
Inter-Symbol Interference MAC

`Z[i] = (1-2X[i]) + (1-2Y[i]) + h*(1-2X[i-1]) + h*(1-2Y[i-1]) + W[i],  h=0.3`

Parameters: h=0.3, SNR=6.0 dB (sigma^2=0.251)
Path: make_path(N, N), corner rate
Design proxy: GMAC Class C at SNR=6 dB
NPD: chained decoder (Stage 1: U from z, Stage 2: V from z,u_hat)

N	ku	kv	Joint Trellis	Chained Trellis	Memoryless SC	NPD (best)	NPD model
16	4	7	0.1660	0.1690	0.1870	0.1380	d16 curric
32	7	15	0.0830	0.0820	0.1130	0.0570	d16 curric
64	15	29	0.0260	0.0410	0.0790	0.0280	d16 h=100
128	30	58	0.0180	0.0220	0.1010	0.0300	d64 h=128
256	59	117	--	0.0060	0.2260	0.0110	d64 h=128
512	119	233	--	0.0030	0.5020	--	--
1024	238	467	--	0.0070	0.8700	--	--

BLER vs N -- NPD beats chained trellis SC by 19-32% at N=16,32,64. At N=64, NPD matches joint trellis SC.



8. Ising MAC Class C (corner rate) -- Memory Channel

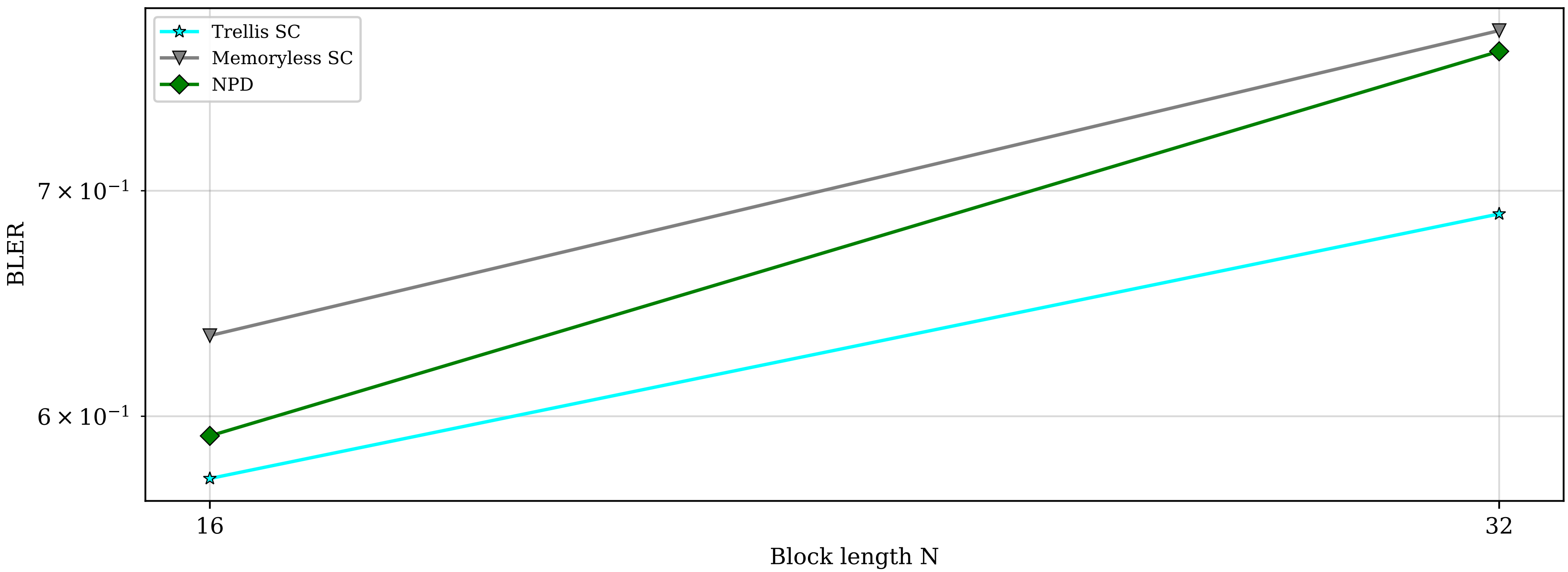
Ising Channel with Markov State

`Good: Z=(1-2X)+(1-2Y)+W,  Bad: Z=W (pure noise),  p_flip=0.1`

Parameters: sigma^2=0.251, p_flip=0.1 (Markov state flip probability)
Path: make_path(N, N), corner rate
Design proxy: GMAC Class C at SNR=6 dB
Extremely hard channel -- BLER > 57% at all tested N

N	ku	kv	Trellis SC	Trellis errs	Memless SC	Memless errs	NPD d=16	NPD errs
16	4	7	0.5750	2873/5k	0.6340	1902/3k	0.5920	2960/5k
32	7	15	0.6890	3443/5k	0.7810	2342/3k	0.7700	3850/5k

BLER vs N -- Ising MAC is extremely hard (BLER>57%). NPD partially learns memory (beats memoryless by 4-2%) but not trellis SC.



9. MA-AGN MAC Class C (corner rate) -- Continuous-State Memory

Moving-Average Additive Gaussian Noise MAC

`Z[i] = (1-2X[i]) + (1-2Y[i]) + N[i],  N[i] = alpha*N[i-1] + W[i],  alpha=0.3`

Parameters: alpha=0.3, sigma^2=0.251 (SNR=6 dB), AR(1) noise process
Path: make_path(N, N), corner rate
Design proxy: GMAC Class C at SNR=6 dB
No analytical trellis exists (continuous state) -- only neural decoder can learn memory

N	ku	kv	Memless SC	Memless errs	NPD (best)	NPD model	NPD/SC
16	4	7	0.1750	349/2k	0.1380	d=32 h=128 BiGRU	0.79x
32	7	15	0.0770	153/2k	0.1120	d=32 h=128 BiGRU	1.46x
64	15	29	0.0250	75/3k	0.0350	d=16 h=100	1.42x
128	30	58	--	--	--	--	--

BLER vs N -- NPD beats memoryless SC at N=16 (21%). At N>=32, chained marginalisation hurts more than memory helps.